Scalability & Future Plan

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction using AI/ML)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Application runs only on a local Flask development server and is not publicly hosted	Evaluators or other users cannot access the application remotely without manual local setup	**High**
2	Model is trained on a single static HDI dataset (one time period) without support for year-wise or real-time updates	Predictions reflect only historical patterns from the training dataset and cannot adapt to newer data automatically	**Medium**
3	No database is used; data and model are stored as flat files (CSV and Pickle)	Limits scalability for handling multiple datasets, user histories, or concurrent large-scale data operations	**Medium**

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single-user, local server handling one request at a time	Deploy on a cloud platform (Render, Heroku, or AWS) with a production WSGI server (Gunicorn) to support multiple concurrent users
2	**Data Storage**	Static CSV file and Pickle model stored locally in project folders	Migrate to a relational database (PostgreSQL/MySQL) for the dataset and use cloud storage (AWS S3) for model artifacts
3	**Performance**	Model loads into memory at server	Implement model caching and consider a

		startup; adequate for single-user local testing	lightweight model-serving layer (e.g., Flask-Caching or a dedicated inference API) for faster response under higher load
4	**Security**	No authentication; open access to all routes since the app runs locally only	Add user authentication (login system) and HTTPS enforcement if the application is deployed publicly

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the Flask application to a cloud hosting platform with a public URL	1–2 months	Makes the application accessible to evaluators and users without local setup
Phase 2	Integrate a relational database to store HDI datasets across multiple years and support historical trend predictions	3–4 months	Enables year-wise predictions and richer historical analysis beyond a single static dataset
Phase 3	Expand the model to support additional indicators (e.g., inequality-adjusted HDI, gender development index) and experiment with advanced models (Random Forest, XGBoost) for improved accuracy	5–6 months	Improves prediction accuracy and broadens the scope of development insights offered
Phase 4	Add user authentication, prediction history tracking, and an admin panel for dataset management	7–9 months	Transforms the tool into a multi-user platform suitable for research teams and institutions